

S2M-Trek: From Single to Multi-Sphere Transport via Per-Frame Deep Sets on a Wheel-Legged Robot

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Abstract: Conventional approaches to object transport typically rely on external restraints such as containers, fences, or tethers to secure the payload and prevent it from falling during transit. In this work, we study the problem of scaling dynamic loco-manipulation from a single free-rolling sphere to multiple spheres transported simultaneously on the back of a wheel-legged robot, without fences, grippers, or mechanical stops. Multiple identical free-rolling spheres form an unordered set with no persistent identity, their ordering may change independently at each history frame, creating a **per-frame permutation symmetry**. Standard history-concatenation set encoders do not explicitly enforce—they impose only a shared, diagonal permutation symmetry over the full history. We show that this symmetry mismatch leads to a concrete failure mode in curriculum-based reinforcement learning. Within the same PPO training budget, flat MLPs and branch-wise encoders plateau at or below the two-sphere stage, while a history-concatenation Deep Sets baseline (DSHC) fails to progress past the two-sphere stage in the submitted run unless ball-to-slot assignments are randomized during training, suggesting that it can exploit slot indices as a curriculum shortcut rather than learning identity-free multi-sphere dynamics; a seed study shows that this early plateau is not a hard architectural barrier, but additional seeds advance late and reach lower strict success. We propose **Per-Frame Deep Sets (PFDS)**, which performs permutation-invariant pooling within each history frame before temporal readout; we prove that PFDS is G_{frame} -invariant and universally approximates continuous G_{frame} -invariant policies. A 2×2 ablation over encoder architecture and slot randomization separates the architectural and data-augmentation pathways, and PFDS reaches the five-sphere stage with 100% no-drop transport in simulation across all five random seeds. We further distill the PFDS teacher into TACTSET via DAGger, replacing privileged sphere-state observations with a compact binary union contact map that is naturally G_{frame} -invariant. Finally, the same deployment stack runs closed-loop on the physical wheel-legged robot at every sphere count $k=1, \dots, 5$, and measured contact maps show connected-component counts matching the sphere counts while carrying no persistent object identity; these hardware results are qualitative and complement the quantitative simulation study.

Keywords: locomotion, loco-manipulation, permutation invariance, deep sets, teacher-student learning, tactile sensing, multi-object manipulation

1 Introduction

Wheel-legged and legged robots have achieved remarkable locomotion versatility [2, 3, 4, 5], and recent work has extended these platforms to single-object loco-manipulation [6, 7, 8, 9]. Scaling from one object to many, however, is not a simple matter of adding more input channels. Transporting *multiple* free-rolling objects simultaneously—without fences, clamps, or fixed identity labels—

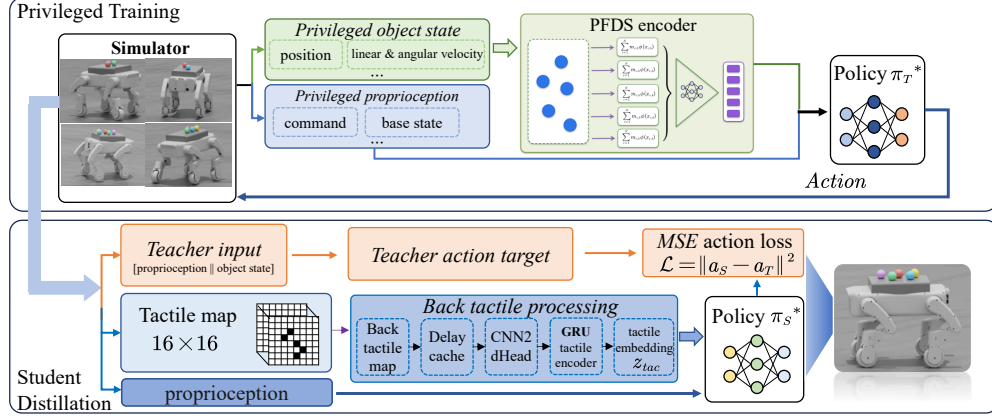


Figure 1: System overview of S2M-Trek (**top**: privileged teacher training; **bottom**: tactile-student distillation pipeline). The PFDS teacher is trained in Isaac Lab [1] with PPO using privileged object-state observations (position, linear/angular velocity). TACTSET is distilled via DAgger, replacing the privileged observation with a binary union tactile contact map; the same deployment stack is validated qualitatively on the physical robot in Section 5.5.

changes the representational problem: the objects form an unordered, identity-free set whose slot assignments may change independently from one history frame to the next, a symmetry that prior work has not formalised. We study this problem with large-scale simulation and qualitative physical validation: all quantitative comparisons are in simulation, while the physical wheel-legged platform provides closed-loop feasibility tests and measured tactile maps.

The symmetry mismatch problem. Consider the multi-sphere observation tensor $X \in \mathbb{R}^{H \times N \times d}$, where H is the history length, N the maximum number of object slots, and d the per-object feature dimension. Because the spheres are identical, swapping two balls’ slot assignments in any single history frame produces a physically equivalent state. The correct quotient group is therefore the *per-frame* product group

$$G_{\text{frame}} = (\mathfrak{S}_N)^H, \quad (g \cdot X)_{t,i} = x_{t,\pi_t^{-1}(i)}, \quad g = (\pi_1, \dots, \pi_H), \quad (1)$$

not the *diagonal* subgroup $G_{\text{diag}} = \{(\pi, \dots, \pi) : \pi \in \mathfrak{S}_N\} \subsetneq G_{\text{frame}}$ that history-concatenation encoders [10, 11] actually satisfy. The quotient ratio $|G_{\text{frame}}|/|G_{\text{diag}}| = (N!)^{H-1}$ reaches $\approx 1.7 \times 10^6$ for $N=5, H=4$: six orders of magnitude of redundant non-physical variation that G_{diag} encoders cannot collapse.

Two paths to G_{frame} robustness. One can approach G_{frame} invariance in two ways:

1. **Architectural (intrinsic):** Design the encoder so that $f(g \cdot X) = f(X)$ for all $g \in G_{\text{frame}}$ by construction.
2. **Data-augmentation (learned):** Keep a G_{diag} encoder but sample a fresh per-frame slot permutation π_t at every simulation step, implicitly averaging the loss over the G_{frame} orbit.

We show experimentally that these two paths are not interchangeable: the architectural path advances the curriculum in both training regimes, while in the submitted run the data-augmentation path fails to advance past the two-sphere stage when augmentation is absent. Additional seeds show that this is a slow, seed-dependent curriculum effect rather than a hard ceiling.

Contributions.

1. We formalise multi-sphere transport as a G_{frame} -invariant policy learning problem, deriving the quotient ratio $(N!)^{H-1}$ as a quantitative measure of the representational gap left unaddressed by standard encoders.

2. We propose **PFDS** (Per-Frame Deep Sets): a minimal modification of standard Deep Sets that achieves G_{frame} -invariance by performing frame-wise pooling before temporal readout, backed by proofs of invariance and universal approximation.
3. We present an eight-encoder comparison and a 2×2 ablation ($\text{encoder} \in \{\text{PFDS}, \text{DSHC}\} \times \text{augmentation} \in \{\text{on}, \text{off}\}$) that separates the architectural and data-augmentation contributions, and we further confirm PFDS’s reliability across five random seeds. We also retrain DSHC_OFF from additional seeds and show that the submitted early plateau is seed-dependent rather than a hard architectural barrier: without augmentation the curriculum is slow, seed-dependent, and low-yield.
4. We propose TACTSET, a DAgger-distilled student that replaces privileged ball-state observations with a compact binary union contact map generated from a 16×16 array—a representation that is *naturally* G_{frame} -invariant—achieving 75% no-drop transport of five spheres in simulation.
5. We run the deployment stack closed-loop on the physical wheel-legged robot at every sphere count $k=1, \dots, 5$ and record the binary contact maps from a 16×16 pressure array. The measured maps contain one connected contact component per sphere but no persistent identity, directly instantiating the identity-free observation model that motivates per-frame invariance. These hardware results are qualitative; our quantitative claims remain simulation-based.

2 Related Work

Legged loco-manipulation. Rapid motor adaptation [5] and massively parallel RL [3] laid the foundation for agile locomotion. DribbleBot [6] and follow-up work [7, 8] extended quadrupeds to single-object manipulation, and LocoTouch [9] added back-mounted tactile sensing for cargo transport. To our knowledge, no prior work addresses unconstrained multi-sphere transport or identifies per-frame permutation symmetry as a representational bottleneck for scaling loco-manipulation.

Permutation-invariant networks. Deep Sets [10] introduced the $\rho(\sum_i \phi(x_i))$ universal approximator; Set Transformer [11] and PointNet [12] extended it with attention and point-cloud structure, and Maron *et al.* [13] analysed higher-order tensor invariance. These methods all target *single-frame* sets. We address the strictly harder *multi-frame* case, where each history frame may be permuted independently—a product group $(N!)^{H-1}$ larger than the diagonal group these encoders satisfy.

Equivariant RL and multi-object symmetry. MDP homomorphic networks [14, 15], SE(3)-equivariant manipulation [16, 17], and multi-agent symmetry methods [18] exploit spatial or agent-level structure for sample-efficient RL. None considers the temporal product symmetry arising when identical objects are observed over multiple history frames with independent slot re-assignments.

Tactile sensing and distillation. Tactile sensors support contact-rich manipulation and load transport on legged platforms [19, 20, 21, 22, 9], and DAgger [23] distillation bridges privileged training observations and deployment-feasible sensors. A central obstacle in multi-object settings is that object-indexed teacher representations do not map cleanly to tactile signals; we observe that the Boolean *union* of contact footprints is G_{frame} -invariant by construction and therefore aligns student and teacher representations without explicit permutation bookkeeping.

3 Problem Formulation

Multi-sphere transport MDP. A four-wheel-independent-steering quadruped—self-designed for this study—transports $k \in \{1, \dots, N\}$ identical free-rolling steel spheres (radius 2 cm, diameter 4 cm) on a flat back plate ($22.4 \text{ cm} \times 15.1 \text{ cm}$). $N = 5$; the robot has 12 leg joints and 4 hub motors, controlled at 50 Hz. A back plate equipped with a 16×16 binary tactile array (256 cells) provides contact feedback. The episode terminates when any ball falls off the plate or the robot falls. Object count k is increased by a multi-criterion curriculum (episode-length ratio, support margin, dangerous fraction, edge-overflow fraction, velocity tracking error).

Multi-ball observation. At policy time t the agent observes a proprioception window $p_t \in \mathbb{R}^{d_p}$ (history length 6, see Section C.2) and a multi-ball history tensor $X_t \in \mathbb{R}^{H \times N \times d}$ ($H=4$ object frames, $d=14$: ball position, velocity, orientation, angular velocity, activity flag). Inactive slots are zero-masked. We use H exclusively for the object-history length throughout the main body, and write X for a generic such tensor when no time index is needed. Because the balls are identical, the correct physical equivalence class of a history tensor X is

$$[X]_{G_{\text{frame}}} = \{g \cdot X : g \in G_{\text{frame}}\},$$

and the policy value function should be constant on each orbit. We define the *curriculum-reachable ball count* at training budget B :

$$K_{\max}(E; B) = \max\{k : \text{Succ}_k(E; B) \geq \eta\}, \quad (2)$$

where Succ_k is the success rate on k -ball evaluation episodes and η is the advancement threshold.

Evaluation metrics. We report two post-training success metrics, evaluated over 100 episodes per sphere count. Teacher policies are evaluated for 1000 steps (20 s); the distilled student is evaluated for 500 steps (10 s), matching the horizon used during its distillation. The 95% Wilson confidence interval is approximately ± 5 –8 pp at success rates near 0 or 1 and widest ($\approx \pm 10$ pp) near 50%. *No-drop*: the episode completes the full horizon without any sphere falling off the plate—this is the *primary* task metric, measuring whether the symmetry-aware encoder can maintain multi-sphere transport. *Strict*: additionally requires zero near-edge events throughout the episode (the ball never exits the safe zone), measuring conservative margin performance. Because near-edge events are sensitive to the amount of training received at each curriculum level, strict success conflates architectural quality with training exposure; we therefore report both but treat no-drop as the primary measure of encoder capability.

Physical evaluation. Hardware runs use the same wheel-legged platform and deployment stack as the final system. We run closed-loop trials at every sphere count $k=1, \dots, 5$ and record synchronized video and the binary contact map. Because repeating matched initial conditions at scale is difficult on hardware, these runs are used for qualitative feasibility and sensing validation; all quantitative success-rate comparisons remain simulation-based.

4 Method

4.1 Per-Frame Deep Sets (PFDS)

PFDS decouples *within-frame* set aggregation from *cross-frame* temporal fusion. It modifies standard Deep Sets [10] as

$$h_t = \text{Pool}(\{\phi(x_{t,i})\}_{i \in \mathcal{A}_t}), \quad t = 1, \dots, H, \quad (3)$$

$$z = \rho([h_1; h_2; \dots; h_H]), \quad (4)$$

where $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^e$ is a shared per-element MLP, Pool is a continuous permutation-invariant aggregator on finite multisets (Definition 1; we use masked sum throughout, and the construction also admits mean or max), and ρ is a temporal readout MLP. A slot permutation π_t acts on both the observation tensor and the activity mask, mapping $(\mathcal{A}_t, x_{t,\cdot})$ to $(\pi_t(\mathcal{A}_t), x_{t,\pi_t^{-1}(\cdot)})$, so that h_t depends only on the multiset $\{x_{t,i}\}_{i \in \mathcal{A}_t}$.

Proposition 1 (G_{frame} -invariance of PFDS). f_{PF} is G_{frame} -invariant in the sense of Definition 2: for every $g \in G_{\text{frame}}$ and every X , $f_{\text{PF}}(g \cdot X) = f_{\text{PF}}(X)$.

(Proof in Section A.)

Contrast with DSHC. The standard Deep Sets encoder [10] applied to multi-frame histories forms per-slot tokens $y_i = [x_{1,i}; \dots; x_{H,i}] \in \mathbb{R}^{Hd}$ and computes

$$f_{\text{HC}}(X) = \rho\left(\sum_{i=1}^N \phi_Z(y_i)\right), \quad \phi_Z : \mathbb{R}^{Hd} \rightarrow \mathbb{R}^e. \quad (5)$$

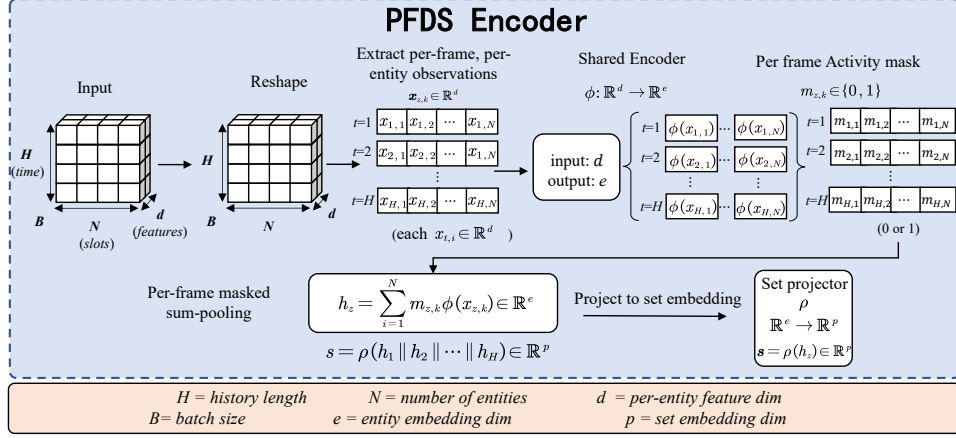


Figure 2: PFDS encoder architecture. Per-object observations $x_{t,i}$ are pooled independently within each history frame t to produce frame embeddings h_t ; these are then concatenated and processed by the readout MLP ρ . Because pooling occurs *within* each frame, the encoder is invariant to independent per-frame permutations (G_{frame} -invariant by Proposition 1).

We call this DSHC (Deep Sets, History-Concat). Because ϕ_Z couples frame index t with slot index i , f_{HC} satisfies only G_{diag} invariance, not G_{frame} . PFDS recovers G_{frame} invariance by pooling within each frame *before* temporal concatenation.

Proposition 2 (G_{diag} but not G_{frame} invariance of DSHC). f_{HC} is G_{diag} -invariant. It is, however, not G_{frame} -invariant in general (in the sense of Definition 2): for any $H \geq 2$ and $N \geq 2$ there exist a continuous feature map ϕ_Z (for instance the flattened outer product $\phi_Z(y) = \text{vec}(yy^T)$), a continuous readout ρ , an input X , and an element $g \in G_{\text{frame}} \setminus G_{\text{diag}}$ such that $f_{\text{HC}}(g \cdot X) \neq f_{\text{HC}}(X)$.

(Proof in Section A.)

The ratio $|G_{\text{frame}}|/|G_{\text{diag}}| = (N!)^{H-1}$ quantifies the residual representational gap; for $N=5$, $H=4$ this equals $(120)^3 \approx 1.7 \times 10^6$.

Theorem 1 (Universal approximation of PFDS for G_{frame} -invariant functions). Fix the per-frame cardinalities $m_t = |\mathcal{A}_t|$ for $t = 1, \dots, H$ and let $K \subset \mathbb{R}^d$ be a compact box of admissible per-object observations. Let $\Omega = \prod_{t=1}^H K^{m_t}$, and let $G = \prod_{t=1}^H \mathfrak{S}_{m_t}$ act on Ω frame-wise by permuting each K^{m_t} factor. This action coincides with the induced action of G_{frame} on the fixed-cardinality canonical slice $\Omega \hookrightarrow \mathbb{R}^{H \times N \times d}$ (active-slot embedding with the remaining $N - m_t$ slots zero-masked). If $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^e$ and $\rho: \mathbb{R}^e \rightarrow \mathbb{R}^p$ are realised by sufficiently wide continuous MLPs and Pool is sum-pooling, then PFDS is dense, under the sup-norm, in the space of continuous G -invariant functions $f: \Omega \rightarrow \mathbb{R}^p$ (equivalently, in the continuous functions on the quotient $\mathcal{X} = \Omega/G$).

(Proof sketch in Section A; the result is a direct corollary of Deep Sets universality applied frame-wise followed by an MLP readout.)

4.2 Tactile Student Policy (TactSet)

The PFDS teacher uses privileged ball-state observations X_t during training. On the physical robot these are unavailable; only a binary contact map from an array sensor is observed. TACTSET bridges this gap by imitating the teacher from a tactile map $\tau_t \in \{0, 1\}^{R \times C}$ defined as the Boolean union of active balls' contact footprints. Let $(\bar{x}_{t,i}, \bar{y}_{t,i}, \bar{z}_{t,i})$ denote the centre of ball i at time t expressed in the back-plate frame, and $(\bar{x}_s, \bar{y}_s)$ the planar centre of cell s in the same frame:

$$\tau_{t,s} = \bigvee_{i \in \mathcal{A}_t} \mathbf{1}(|\bar{x}_{t,i} - \bar{x}_s| < r + \delta_x, |\bar{y}_{t,i} - \bar{y}_s| < r + \delta_y, |\bar{z}_{t,i} - (z_{\text{plate}} + r)| < \epsilon_z), \quad (6)$$

where z_{plate} is the height of the plate surface and the z -condition gates contact between the sphere's bottom pole and the plate. In simulation we rasterize the map on a 21×15 grid matched to the

simulated sensor footprint; the physical array is a 16×16 M1616M sensor, and the deployment stack consumes the same binary-map interface. Because $\vee$ is symmetric in its arguments, τ_t depends only on the multiset of contact footprints and is a function of $[X_t]_{G_{\text{frame}}}$, semantically aligning the student input with the teacher’s observation.

The student receives a two-channel tensor $s_t = [\tau_t; \tau_{t-\delta_t}]$, where δ_t is a per-episode sensing delay sampled uniformly from $\{1, 2\}$ frames during training. This input is processed by a compact CNN pre-encoder (embedding dimension 64) followed by a GRU (hidden size 512); the resulting tactile embedding is concatenated with the proprioception embedding and fed to the policy head (full hyperparameters in Table 4). The student minimises

$$\min_{\theta} \mathbb{E}_{(p_t, X_t, s_t) \sim \mathcal{D}} \left[\left\| \pi_S^{\theta}(p_t, s_t) - \pi_T(p_t, X_t) \right\|_2^2 \right], \quad (7)$$

with DAgger-style online aggregation (behaviour-cloning initialisation followed by 7 rounds of online rollout with PFDS labelling). This two-channel delay representation is part of the deployed pipeline; the hardware sensor itself publishes a binary 16×16 frame at 10 Hz with a 5 g threshold.

5 Experiments

5.1 Setup

All teacher policies are trained in Isaac Lab [1] with PPO [24] using 4096 parallel environments for 30 000 iterations. Reward, curriculum, and hyperparameters are identical across encoders; only the set-encoding module varies. Full hyperparameters are listed in Table 3.

Physical platform. The robot is a custom four-wheel-independent-steering quadruped with 12 leg joints and 4 hub motors, controlled at 50 Hz. Its back plate is covered by a silicone layer over a compliant foam base and a 16×16 M1616M binary pressure array sampled at 10 Hz and thresholded at 5 g. The deployment stack runs the trained policy closed-loop at sphere counts $k=1, \dots, 5$; a controlled sweep synchronized video with the sensor stream (alignment residual below 0.5 s) and recorded 110–119 s per condition. These hardware trials are qualitative and are not used to estimate success rates.

Encoder baselines. We compare eight architectures spanning three symmetry levels (Table 1), all configured to comparable parameter budgets through matched hidden widths. G_0 (*no invariance*): FLAT-MLP flattens X into a single vector; BRANCH-MLP encodes each slot $X_{:,i,:}$ with a shared MLP and concatenates the results. G_{diag} (*invariance*): DSHC [10] forms history-concatenated tokens $y_i = [x_{1,i}; \dots; x_{H,i}]$ and applies Deep Sets pooling; Set Transformer [11] replaces the inner network with ISAB/PMA attention on the same tokens. G_{frame} (*invariance (ours)*): PFDS pools within each history frame before temporal readout (Proposition 1). Set Transformer (PF) is the higher-capacity analogue—frame-wise ISAB/PMA attention—but is not used as the distillation teacher because its wall-clock training time is the highest among G_{frame} encoders. PFDS_OFF disables slot randomisation for PFDS, isolating the architectural guarantee from the data-augmentation pathway.

5.2 Main Comparison: Architecture Determines Curriculum Reachability

Figure 4 reports curriculum progression and Figure 3 reports per-encoder success rates. The submitted runs form a clear hierarchy consistent with symmetry-group containment: G_0 encoders (FLAT-MLP, BRANCH-MLP) stall at or below $k=2$; G_{diag} encoders (DSHC, Set Transformer) reach $k=5$ with slot-permutation augmentation, while the submitted DSHC_OFF run (augmentation off) stalls at $k=2$; G_{frame} encoders (PFDS, Set Transformer (PF)) advance under both regimes, the architectural guarantee that motivates our construction. The DSHC_OFF row is seed-dependent, however: additional seeds eventually advance (but late and to lower strict success), as we quantify in Section 5.4 and Section F. The accurate conclusion is therefore that without a G_{frame} -invariant encoder, curriculum progression is slow, seed-dependent, and low-yield—not architecturally impossible.

Table 1: Teacher encoder comparison under equal PPO budget (30 000 iters). $K_{\max}$: highest curriculum level reached. “Adv. to k ”: iterations (thousands) to first reach that level; “—” = not reached within budget. Inv.: invariance group. Strict/no-drop: 5-ball success rates (100 trials, 95% Wilson CI $\approx \pm 5-8$ pp); no-drop is the primary metric (Section 3). The DSHC.OFF row is the submitted seed; additional seeds and the resulting seed-dependence are reported in Section F. †: only G_{frame} encoder reaching $k=5$ without slot-permutation augmentation.

Encoder	Inv.	perm	$K_{\max}$	Adv. to 3	Adv. to 5	strict	no-drop
FLAT-MLP	G_0	T	≤ 1	—	—	0%	0%
BRANCH-MLP	G_0	T	≤ 2	—	—	0%	2%
DSHC.OFF	G_{diag}	F	≤ 2	—	—	0%	0%
DSHC	G_{diag}	T	5	$\sim 7.4\text{k}$	$\sim 10.9\text{k}$	68%	98%
Set Transformer	G_{diag}	T	5	$\sim 3.6\text{k}$	$\sim 7.7\text{k}$	10%	94%
Set Transformer (PF, ours)	G_{frame}	T	5	$\sim 7.5\text{k}$	$\sim 16.4\text{k}$	54%	100%
PFDS.OFF (ours) [†]	G_{frame}	F	5	$\sim 16.9\text{k}$	$\sim 27.9\text{k}$	41%	98%
PFDS (ours)	G_{frame}	T	5	$\sim 3.9\text{k}$	$\sim 6.3\text{k}$	41%	100%

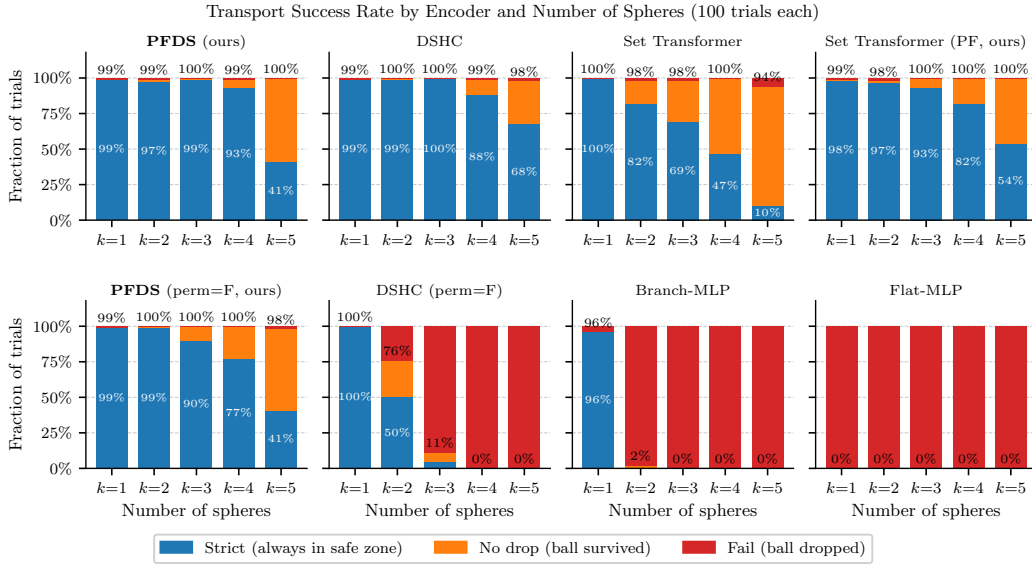


Figure 3: Transport success rates for all eight encoder architectures over 100 trials $\times$ 5 ball counts. Blue: strict (always within safe zone); orange: no-drop (ball survived, may briefly exit); red: fail. G_{frame} -invariant encoders (PFDS, Set Transformer (PF)) achieve 100% no-drop at $k=5$; DSHC with augmentation achieves 98%. G_0 encoders drop to near-zero no-drop beyond $k=1$; the submitted DSHC.OFF run stalls beyond $k=2$ (additional seeds are discussed in Section F).

Primary metric (no-drop). PFDS and Set Transformer (PF) achieve 100% no-drop at $k=5$, the highest of any encoder, and PFDS is the fastest to reach the top curriculum level ($\sim 6.3\text{k}$ iters to $k=5$, vs. $\sim 10.9\text{k}$ for DSHC and $\sim 7.7\text{k}$ for Set Transformer). The 2 pp no-drop gap over DSHC is within sampling noise at 100 trials; the substantive differentiator is that PFDS.OFF still attains 98% no-drop without augmentation while the submitted DSHC.OFF run does not advance past $k=2$ within budget. Robustness with a G_{frame} -invariant architecture is therefore independent of data augmentation in our runs, whereas with DSHC it is not.

5.3 2×2 Key Ablation: Two Paths to G_{frame} Robustness

Table 2 and Figure 4 show the central empirical pattern: in the submitted runs DSHC reaches $k=5$ only with per-step augmentation ($\pi_t \sim \text{Uniform}(\mathfrak{S}_N)$) and stalls at $k=2$ without it within the 30 000-iteration budget—slot index becomes a spurious proxy for curriculum stage that the G_{diag} encoder cannot ignore. PFDS succeeds in both settings, and the additional-seed results below show

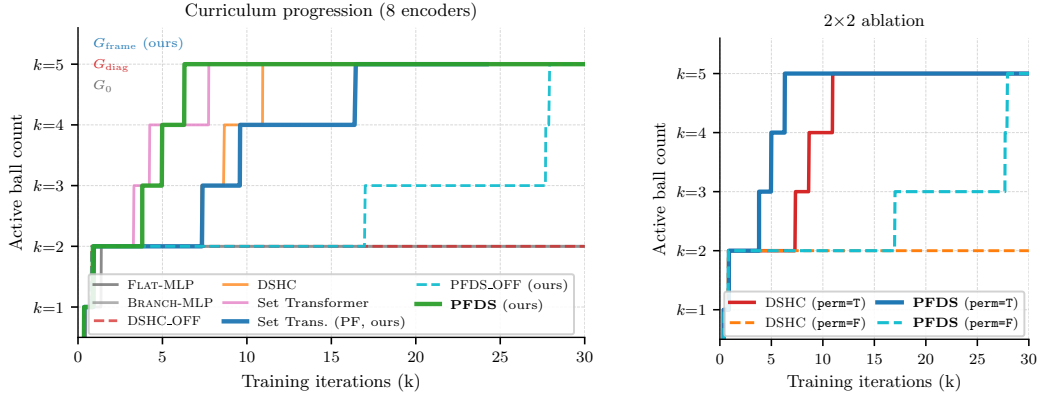


Figure 4: Training dynamics across encoder variants. **Left:** curriculum progression (active ball count vs. iteration) for all eight encoders. G_{frame} architectures (PFDS, Set Transformer (PF)) reach $k=5$ reliably; G_{diag} encoders only with slot-permutation augmentation; G_0 encoders stall at $k \leq 2$. **Right:** 2×2 ablation isolating the architectural and data-augmentation pathways. PFDS reaches $k=5$ with and without augmentation; the submitted DSHC run stalls at $k=2$ without it.

Table 2: 2×2 ablation: encoder $\times$ per-step orbit augmentation. “Adv. to 3” is the number of training iterations at which the curriculum first reaches $k \geq 3$; “—” means not reached by 30 000 iterations.

Encoder	permute=True		permute=False	
	K_{max}	Adv. to 3	K_{max}	Adv. to 3
DSHC (G_{diag})	5	$\sim 7.4\text{k}$	≤ 2	—
PFDS (G_{frame})	5	$\sim 3.9\text{k}$	5	$\sim 16.9\text{k}$

that the augmentation-off failure is a budget- and seed-limited curriculum effect rather than a capacity ceiling.

5.4 Post-Review Robustness, Diagnostics, and Distillation

Post-review evidence. We summarize the new experiments here; full tables and measurement protocols are in [Section F](#).

Seed dependence. The submitted DSHC_OFF seed stalls at $k=2$, but two additional seeds eventually reach $k=5$ (one at $\sim 15\text{k}$ and one at $\sim 22\text{k}$ iterations) while sitting at $k=2$ through most of the budget. Under the paper protocol their strict success at $k=5$ is only 22–23% (no-drop 87–100%), versus 39–41% strict / 98–100% no-drop for PFDS. Without augmentation the curriculum is slow, seed-dependent, and low-yield—not architecturally impossible.

Slot shortcut and invariance. Applying a fresh per-frame relabeling to DSHC_OFF at $k=2$ reduces strict success from 91% to 53% (−38 pp), while PFDS and DSHC remain within sampling noise. At the action level, PFDS deviates by $3.6\text{--}6.4 \times 10^{-4}$ of action std under G_{frame} permutations—numerical-noise level—whereas augmentation-trained DSHC deviates by 9.5–20%, $\sim 265\text{--}315 \times$ larger, with the largest gap in low-margin states.

Capacity versus curriculum. Trained directly at $k=5$ with no curriculum, both encoders learn transport (no-drop 98% for DSHC_OFF, 100% for PFDS_OFF; strict 73%/70% and 21%/32% under permute-on/off), so the plateau is not an architectural capacity limit. These fixed- k specialists score 0% at every $k < 5$, however, whereas the curriculum yields one policy for all counts.

TACTSET robustness and distillation. Re-evaluated in our harness (baseline 85% no-drop at $k=5$), 1–10% taxel dropout costs at most 1 pp, 20% dropout or spurious contacts degrade gracefully, row/column calibration shifts reduce no-drop to 86%/61%, and latency is the dominant sensitivity (71% at +1 frame, 48% at +3). Pure behavior cloning reaches only 4% no-drop at $k=5$, one on-policy DAgger round lifts this to 72%, and the shipped round-7 student measures 85% in the same harness;

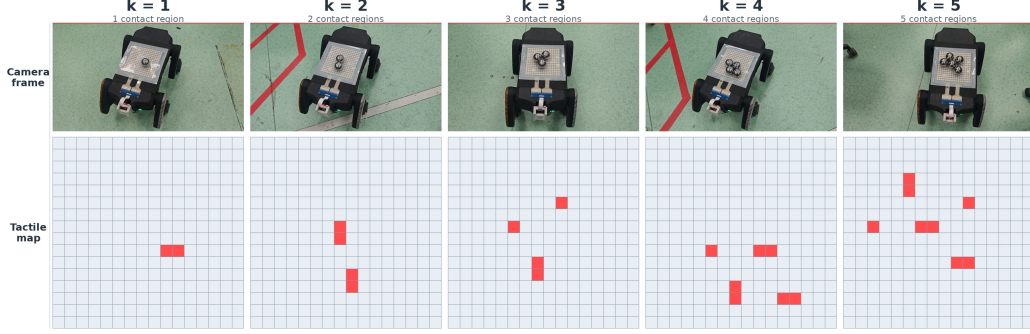


Figure 5: Physical validation on the wheel-legged platform. Top: video frames of controlled closed-loop runs with $k=1, \dots, 5$ steel spheres. Bottom: the binary 16×16 contact map measured at the same instant (threshold 5 g; no interpolation). At the selected frame of each condition, the map produces one 8-connected contact component per sphere, but carries no identity: which sphere generated which component cannot be recovered from a single frame. These are qualitative single runs; no hardware success statistic is claimed.

most of the gain is therefore on-policy, and the residual gap is representation-limited (velocity and per-object attribution discarded by the binary map).

Transport quality. At $k=5$, PFDS keeps all spheres on the plate in 99% of episodes, but the median episode’s minimum margin touches the safe-zone boundary; all 59 strict failures are transient near-edge excursions and none is a drop (mean near-edge dwell 0.1%). Strict success therefore measures dynamic containment margin rather than transport capability. With $r=2$ cm, static packing is not the binding constraint; stronger margin objectives trade off against velocity tracking, whose error grows from 0.059 m/s at $k=4$ to 0.068 m/s at $k=5$.

5.5 Physical Validation

We ran the deployed stack closed-loop at every sphere count $k=1, \dots, 5$ on the physical robot. For each count we recorded at least one controlled run with synchronized video and the measured 16×16 binary contact map; Figure 5 shows a representative frame and the same-instant map. At the selected frame of each condition, the 8-connected component count equals the sphere count, but no component carries identity or persistent labels: a real estimator can count contacts yet must solve per-frame data association for object-indexed states. These are qualitative demonstrations; we report no hardware success statistics.

6 Conclusion

We introduced Per-Frame Deep Sets (PFDS), an intrinsically G_{frame} -invariant encoder for multi-sphere transport. Aligning the encoder with $G_{\text{frame}} = (\mathfrak{S}_N)^H$ —the diagonal group is smaller by a factor $(N!)^{H-1}$ —is the key design step. The eight-encoder comparison and 2×2 ablation show that G_{frame} -invariant architectures are the only ones robust without slot-permutation augmentation, while additional seeds show that the augmentation-free DSHC failure is a slow, seed-dependent curriculum effect rather than a hard capacity barrier. PFDS reaches $k_{\text{max}}=5$ across five seeds, TACTSET distills it into a tactile student, and the same stack runs closed-loop on hardware at every $k=1, \dots, 5$, where measured contact maps contain one connected component per sphere but no persistent identity.

Limitations and future work. Quantitative evaluation is simulation-based and physical runs are qualitative. The student’s 21×15 map differs from the 16×16 deployed array, so resolution and calibration remain future work. With $r=2$ cm, static packing does not limit $k=5$, and 99% of $k=5$ episodes retain every sphere; strict failures are transient boundary excursions. Stronger margin objectives and explicit object matching remain future work.

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A Proof Details

A.1 Preliminaries: Permutation and G_{frame} -Invariance

For completeness we recall the two notions of invariance used throughout the paper.

Definition 1 (Permutation invariance of a multiset aggregator). *Let $\mathcal{M}_{\leq N}(\mathbb{R}^e)$ denote the set of finite multisets in $\mathbb{R}^e$ of cardinality at most N . A continuous map $\text{Pool} : \mathcal{M}_{\leq N}(\mathbb{R}^e) \rightarrow \mathbb{R}^e$ is permutation-invariant if its value depends only on its argument as a multiset (with multiplicity), i.e. for any finite indexed family $(z_i)_{i \in \mathcal{A}}$ with $|\mathcal{A}| \leq N$ and any bijection $\sigma : \mathcal{A} \rightarrow \mathcal{A}$,*

$$\text{Pool}(\{z_{\sigma(i)}\}_{i \in \mathcal{A}}) = \text{Pool}(\{z_i\}_{i \in \mathcal{A}}).$$

Equivalently, for each cardinality $m \leq N$, the restriction of Pool to multisets of size m corresponds to a symmetric continuous function $(\mathbb{R}^e)^m \rightarrow \mathbb{R}^e$. Standard examples include masked sum, mean, and max.

Definition 2 (G_{frame} -invariance of a function on history tensors). *With $G_{\text{frame}} = (\mathfrak{S}_N)^H$ acting on $X \in \mathbb{R}^{H \times N \times d}$ by $(g \cdot X)_{t,i} = x_{t,\pi_t^{-1}(i)}$ for $g = (\pi_1, \dots, \pi_H)$, a function $f : \mathbb{R}^{H \times N \times d} \rightarrow \mathbb{R}^a$ is G_{frame} -invariant if*

$$f(g \cdot X) = f(X) \quad \text{for all } X \text{ and all } g \in G_{\text{frame}}.$$

A history tensor X and its G_{frame} -orbit $[X]_{G_{\text{frame}}} = \{g \cdot X : g \in G_{\text{frame}}\}$ are identified under any G_{frame} -invariant f , so f factors through the quotient $\mathbb{R}^{H \times N \times d} / G_{\text{frame}}$.

A.2 Proof of PFDS G_{frame} -Invariance

Proof. Per (1), under $g = (\pi_1, \dots, \pi_H) \in G_{\text{frame}}$ the new slot i at frame t holds the old value at slot $\pi_t^{-1}(i)$; equivalently, the old slot j moves to $\pi_t(j)$, so the active set transforms as $\mathcal{A}_t \mapsto \pi_t(\mathcal{A}_t)$. Therefore

$$h_t(g \cdot X) = \text{Pool}(\{\phi(x_{t,\pi_t^{-1}(i)})\}_{i \in \pi_t(\mathcal{A}_t)}) = \text{Pool}(\{\phi(x_{t,j})\}_{j \in \mathcal{A}_t}) = h_t(X),$$

where the second equality reindexes the multiset by $j = \pi_t^{-1}(i)$, which traverses $\mathcal{A}_t$ exactly once. The two multisets $\{\phi(x_{t,j})\}_{j \in \mathcal{A}_t}$ are identical, so Pool , being a function on multisets, returns the same value. Hence $z(g \cdot X) = \rho([h_1; \dots; h_H]) = z(X)$. $\square$

A.3 Proof of HCDS G_{diag} - but not G_{frame} -Invariance

G_{diag} -invariance. For $g = (\pi, \dots, \pi) \in G_{\text{diag}}$, every frame is permuted by the same π , so each per-slot token transforms as $y_i(g \cdot X) = y_{\pi^{-1}(i)}(X)$. The set $\{y_i\}_{i=1}^N$ is therefore permuted as a multiset, and

$$\sum_{i=1}^N \phi_Z(y_i(g \cdot X)) = \sum_{i=1}^N \phi_Z(y_{\pi^{-1}(i)}(X)) = \sum_{i=1}^N \phi_Z(y_i(X)),$$

so $f_{\text{HC}}(g \cdot X) = f_{\text{HC}}(X)$.

Failure of G_{frame} -invariance ($H=2, N=2$ counterexample). Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with rows indexing time and columns indexing slots, and take $g = (\text{id}, (12)) \in G_{\text{frame}} \setminus G_{\text{diag}}$. Then $(g \cdot X)_{1,:} = (a, b)$ and $(g \cdot X)_{2,:} = (d, c)$, so the DSHC tokens are $y_1^g = (a, d)$, $y_2^g = (b, c)$, whereas $y_1 = (a, c)$, $y_2 = (b, d)$. For the linear feature $\phi_Z = \text{id}$, $\sum_i y_i^g = (a + b, c + d) = \sum_i y_i$: the linear feature is

incidentally invariant. For the flattened outer-product feature $\phi_Z(y) = \text{vec}(yy^\top)$, a direct computation gives

$$\sum_i y_i y_i^\top = \begin{pmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{pmatrix}, \quad \sum_i y_i^g (y_i^g)^\top = \begin{pmatrix} a^2 + b^2 & ad + bc \\ ad + bc & c^2 + d^2 \end{pmatrix},$$

so the $(1, 2)$ -entry differs by $(ad + bc) - (ac + bd) = (a - b)(d - c)$, which is nonzero whenever $a \neq b$ and $c \neq d$. Choose ρ to project onto the coordinate of $\text{vec}(\cdot)$ holding the $(1, 2)$ -entry. Then $f_{\text{HC}}(g \cdot X) - f_{\text{HC}}(X) = (a - b)(d - c) \neq 0$, exhibiting the required ϕ_Z, ρ, X , and $g \in G_{\text{frame}} \setminus G_{\text{diag}}$. The construction trivially extends to any $H \geq 2, N \geq 2$ by padding additional frames and additional slots with constants. $\square$

A.4 Proof Sketch: Universal Approximation of PFDS

Let $m_t, K, \Omega, G, \mathcal{X}$ be as in the theorem, and let $n_{\max} = \max_t m_t$. Since each frame has *exactly* m_t active slots, an orbit in Ω/G is identified with a tuple of multisets $(S_1, \dots, S_H)$ with $|S_t| = m_t$ (no cardinality ambiguity arises across frames).

(i) *Per-frame injection by symmetric power sums.* For each fixed m and any compact $K \subset \mathbb{R}^d$, the map

$$\phi_m : \mathbb{R}^d \rightarrow \mathbb{R}^{M_m}, \quad \phi_m(x) = (p_\alpha(x))_{|\alpha| \leq m},$$

collecting all monomials of degree at most m in the d coordinates of x , makes $\Phi_m(S) = \sum_{x \in S} \phi_m(x)$ a continuous injection on multisets of *exactly* cardinality m : the resulting multivariate power-sum symmetric functions determine the elementary multisymmetric polynomials of S via the multisymmetric Newton identities, and these in turn determine S uniquely. (We use that $|S|$ is known at the time of decoding, so the multiset $\{0, \dots, 0\}$ at one cardinality is distinguishable from the same elements at any other cardinality by context.) Taking ϕ to include the components ϕ_m for all $m \in \{m_1, \dots, m_H\}$ (or, equivalently, all $m \leq n_{\max}$) produces a single continuous $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^M$ that is shared across frames, matching the PFDS architecture in (3). Each per-frame embedding $\Phi_t(S_t) = \sum_{x \in S_t} \phi(x)$ is then continuous and injective on $K^{m_t}/\mathfrak{S}_{m_t}$.

(ii) *Joint injection.* Concatenating per-frame embeddings,

$$v(X) = [\Phi_1(X_1); \dots; \Phi_H(X_H)] \in \mathbb{R}^{HM},$$

gives a continuous map $\Omega \rightarrow \mathbb{R}^{HM}$ that factors through $\mathcal{X}$ and is injective on $\mathcal{X}$, since two orbits agree on v iff their per-frame multisets coincide for every t . Since Ω is compact, so is $\mathcal{X}$, and a continuous bijection from a compact space to a Hausdorff space (here $v : \mathcal{X} \rightarrow v(\mathcal{X}) \subset \mathbb{R}^{HM}$) is a homeomorphism.

(iii) *Readout approximation.* A continuous G -invariant function $f : \Omega \rightarrow \mathbb{R}^a$ descends to a unique continuous $\tilde{f} : \mathcal{X} \rightarrow \mathbb{R}^a$, which by (ii) corresponds to a continuous $\tilde{f} = \tilde{f} \circ v^{-1} : v(\mathcal{X}) \rightarrow \mathbb{R}^a$. Because $v(\mathcal{X})$ is compact (hence closed) in the normal space $\mathbb{R}^{HM}$, Tietze's extension theorem (applied coordinate-wise) extends $\tilde{f}$ to a continuous $\hat{f} : \mathbb{R}^{HM} \rightarrow \mathbb{R}^a$. Restricting $\hat{f}$ to a compact neighbourhood of $v(\mathcal{X})$ and applying the universal approximation theorem for multilayer perceptrons [25] yields a continuous $\rho : \mathbb{R}^{HM} \rightarrow \mathbb{R}^a$ such that $\sup_{X \in \Omega} \|\rho(v(X)) - f(X)\| < \varepsilon$ for any prescribed $\varepsilon > 0$. $\square$

B Hyperparameters

C Environment Specification

C.1 Reward Function

The total reward r_t is the sum of object-balancing terms, locomotion terms, and a penalty term:

$$r_t = r_{\text{obj}} + r_{\text{loco}} + r_{\text{penalty}}. \quad (8)$$

Table 3: PPO hyperparameters shared across all teacher encoder experiments.

Parameter	Value	Note
Parallel environments	4096	Isaac Lab
Total iterations	30 000	
Steps per env per iter	24	Adam
Mini-batches	4	
PPO epochs per iter	5	
Learning rate	3×10^{-4}	
Discount γ	0.99	
GAE λ	0.95	
Clip range ϵ	0.2	
KL target	6×10^{-3}	
Entropy coeff.	0.01	
Value loss coeff.	1.0	
Max grad norm	1.0	
Control frequency	50 Hz	
History length H	4	
Max ball count N	5	

Table 4: DAgger distillation hyperparameters (TACTSET). The physical sensor is a 16×16 binary array sampled at 10 Hz; the simulated student described in Section 4.2 consumes a two-channel 21×15 rasterized binary map with a sampled 1–2 frame delay.

Parameter	Value
Rounds	8
BC steps (round 0)	400 000
DAgger steps (rounds 1–7)	200 000 each
Batch size (training)	20 000
Teacher inference batch	2 048
Learning rate	5×10^{-4}
Tactile input shape	(2, 21, 15)
Pre-encoder	CNN, embedding dim 64
Temporal encoder	GRU, hidden size 512
Delay range (training)	1–2 frames
Physical array	16×16 , 10 Hz, 5 g threshold

Object-balancing rewards. Let $\mathcal{A}_t \subseteq \{1, \dots, N\}$ be the set of active ball indices at time t . For each ball $i \in \mathcal{A}_t$, define the *support margin* $\mu_i \in [0, 1]$ as the minimum normalized distance from ball i ’s projected position to the plate boundary (1 = plate center, 0 = edge):

$$r_{\text{margin}} = +3.0 \cdot \frac{1}{|\mathcal{A}_t|} \sum_{i \in \mathcal{A}_t} \mu_i, \quad (9)$$

$$r_{\text{tail}} = +1.25 \cdot \text{CVaR}_{0.4}(\{\mu_i\}_{i \in \mathcal{A}_t}), \quad (10)$$

where $\text{CVaR}_{0.4}$ [26] is the conditional value-at-risk at the 0.4 tail, i.e. the conditional mean of the lowest 40% of margins (a tail-risk reward term incentivising stabilisation of the most precarious ball).

$$r_{\text{edge}} = -15.0 \cdot \frac{1}{|\mathcal{A}_t|} \sum_{i \in \mathcal{A}_t} \mathbf{1}[\mu_i < 0], \quad (11)$$

$$r_{\text{spacing}} = +1.5 \cdot \frac{2}{|\mathcal{A}_t|(|\mathcal{A}_t| - 1)} \sum_{i < j} \min(\|p_i - p_j\|_2, d_{\text{max}}), \quad (12)$$

$$r_{\text{spacing, tail}} = +0.75 \cdot \text{CVaR}_{0.4}(\{\|p_i - p_j\|_2\}_{i < j}), \quad (13)$$

$$r_{\text{dangerous}} = -20.0 \cdot \mathbf{1}[\min_{i \in \mathcal{A}_t} \mu_i < \mu_{\text{thresh}}], \quad (14)$$

where $p_i \in \mathbb{R}^2$ is the ball’s projected position on the plate, d_{max} is a clipping radius, and μ_{thresh} is the dangerous-state threshold (set per curriculum level).

Velocity-related penalties.

$$r_{\text{vel},xy} = -0.05 \cdot \frac{1}{|\mathcal{A}_t|} \sum_{i \in \mathcal{A}_t} \|v_i^{xy}\|_2^2, \quad (15)$$

$$r_{\text{vel},z} = -0.25 \cdot \frac{1}{|\mathcal{A}_t|} \sum_{i \in \mathcal{A}_t} |v_i^z|, \quad (16)$$

where v_i^{xy} and v_i^z are the ball's horizontal and vertical velocity in the robot frame.

Locomotion rewards (from base locomotion policy).

$$r_{\text{track},v} = +1.5 \cdot \exp(-\|\hat{v}^{xy} - v_{\text{cmd}}^{xy}\|_2^2/0.25), \quad (17)$$

$$r_{\text{track},\omega} = +1.25 \cdot \exp(-|\hat{\omega}_z - \omega_{z,\text{cmd}}|^2/0.25), \quad (18)$$

$$r_{\text{height}} = -1.0 \cdot (h - h_{\text{ref}})^2, \quad (19)$$

$$r_{\text{roll,pitch}} = -1.25 \cdot \|\theta_{\text{rp}}\|_2^2, \quad (20)$$

where $\hat{v}^{xy}$ and $\hat{\omega}_z$ are the robot's base velocity, v_{cmd}^{xy} and $\omega_{z,\text{cmd}}$ are the commanded velocities, h is the base height, h_{ref} is the target height, and θ_{rp} are the roll and pitch angles.

C.2 Observation Space

The teacher policy receives the following observations at each timestep:

Proprioception (history length $H_{\text{prop}} = 6$, distinct from the object-history length $H = 4$).

- Base angular velocity $\omega_b \in \mathbb{R}^3$
- Projected gravity vector $g_b \in \mathbb{R}^3$
- Velocity command $[v_x, v_y, \omega_z] \in \mathbb{R}^3$
- Joint positions $q \in \mathbb{R}^{12}$ (relative to default)
- Joint velocities $\dot{q} \in \mathbb{R}^{12}$
- Previous action $a_{t-1} \in \mathbb{R}^{12}$
- Robot base height $h \in \mathbb{R}^1$

Total proprioception dimension: $43 \times H_{\text{prop}} = 258$.

Multi-ball state (object-history length $H = 4$, per-ball feature dim $d = 14$). For each active ball $i \in \mathcal{A}_t$:

- Relative position $\Delta p_i \in \mathbb{R}^3$ (ball position minus robot base, in robot frame; scale 1.0)
- Relative linear velocity $\Delta v_i \in \mathbb{R}^3$ (scale 0.5)
- Relative orientation as quaternion $q_i \in \mathbb{R}^4$ (scale 1.0)
- Relative angular velocity $\omega_i \in \mathbb{R}^3$ (scale 0.25)
- Activity flag $a_i \in \{0, 1\}$ (scale 1.0)

Inactive slots are zero-masked. Observation noise (uniform): position ± 0.01 m, velocity ± 0.2 m/s, orientation ± 0.05 (Euler-angle equivalent), angular velocity ± 0.2 rad/s. Ball state is observable only when the ball is in contact with the robot surface (contact detection threshold $\approx 10^{-8}$ N); otherwise the observation is zero.

The critic (teacher) additionally receives the denoised version of the above (no observation noise added).

C.3 Termination Conditions

An episode terminates (non-timeout) when any of the following occurs:

1. **Ball falls below robot base:** any active ball’s z -position drops below the robot base z -position, $\exists i \in \mathcal{A}_t : p_{i,z} < p_{\text{robot},z}$.
2. **Robot falls:** the projected gravity vector indicates a roll angle $|\arcsin(g_{b,y})| > 90^\circ$.
3. **Base height too low:** the robot base height falls below the minimum standing height threshold.

Note: the `base_contact` termination (body contact with the ground) is disabled for the multi-sphere task to prevent false positives from the tactile plate surface.

D Curriculum Design

Curriculum learning [27, 28] progresses the agent from easy to hard task instances. In our setting the difficulty axis is the active ball count k .

D.1 Promotion Criteria

The multi-ball curriculum promotes from k to $k + 1$ active balls when a rolling window of recent episodes satisfies *all six* of the following conditions simultaneously:

1. **Episode-length ratio** ≥ 0.85 : the mean episode length divided by the maximum episode length exceeds 85%, indicating the agent consistently survives to episode end.
2. **Support margin** $\geq \mu_k^*$: mean minimum support margin across active balls exceeds the level threshold.
3. **Dangerous fraction** $\leq f_{d,k}^*$: the fraction of timesteps with any ball in a dangerous state (near edge) is below the level threshold.
4. **Edge-overflow fraction** $\leq f_{e,k}^*$: the fraction of timesteps with any ball outside the plate boundary is below the level threshold.
5. **Linear velocity tracking error** $\leq \epsilon_{v,k}^*$: mean $\|\hat{v}^{xy} - v_{\text{cmd}}^{xy}\|_2$ is below threshold.
6. **Angular velocity tracking error** $\leq \epsilon_{\omega,k}^*$: mean $|\hat{\omega}_z - \omega_{z,\text{cmd}}|$ is below threshold.

All conditions must hold simultaneously for `required_successes = 1` consecutive evaluation windows, with `min_stage_episodes = 2048` and `min_steps_per_level = 3200`.

D.2 Per-Level Promotion Thresholds

Table 5: Per-level curriculum promotion thresholds ($k \rightarrow k + 1$). “Support margin” μ^* is the normalized minimum distance from plate boundary. “Dangerous frac.” f_d^* and “Edge frac.” f_e^* are allowed fractions of unsafe timesteps. “Vel. err.” ϵ_v^* and “Yaw err.” ϵ_ω^* are maximum allowed tracking errors.

Level ($k \rightarrow k + 1$)	μ^*	f_d^*	f_e^*	ϵ_v^*	ϵ_ω^*
1 $\rightarrow$ 2	0.30	0.20	0.02	0.30	0.40
2 $\rightarrow$ 3	0.25	0.25	0.03	0.35	0.45
3 $\rightarrow$ 4	0.20	0.30	0.04	0.40	0.50
4 $\rightarrow$ 5	0.15	0.35	0.05	0.45	0.55

The thresholds are relaxed progressively at higher levels to account for the increased difficulty of balancing more balls simultaneously.

E Discussion

Architectural vs. augmentation guarantees. In the submitted run, the failure of `DSHC_OFF` behaves like a budget-limited barrier: slot identity becomes a deterministic proxy for curriculum stage and blocks promotion beyond $k=2$ for the full 30 000-iteration budget. The additional-seed study in Section F shows that this is not irreversible—some seeds do eventually advance—but they do so late in training and reach substantially lower strict success. The architectural reading is therefore

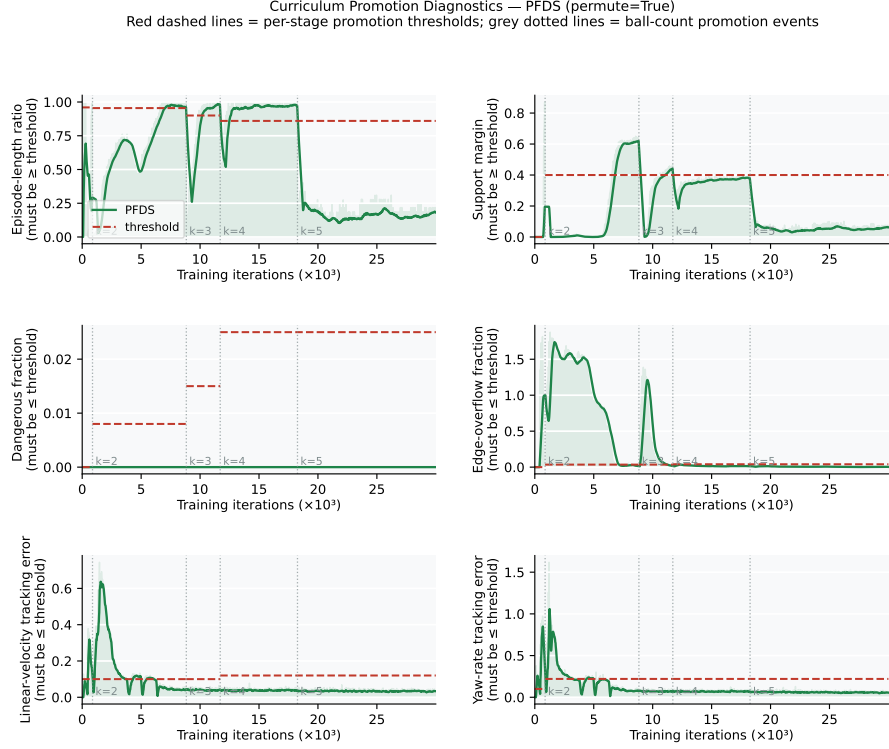


Figure 6: Curriculum promotion-criterion traces for a representative PFDS training run. All six promotion conditions are shown: episode-length ratio, support margin, dangerous-state fraction, edge-overflow fraction, linear-velocity tracking error, and yaw-rate tracking error. Red dashed lines are per-stage promotion thresholds; grey dotted lines mark the iterations at which the curriculum advances to the next level. All six conditions must simultaneously meet threshold for promotion.

conservative: without a G_{frame} -invariant encoder, curriculum progression is slow, seed-dependent, and low-yield. PFDS avoids this by construction; without augmentation it advances to $k=5$ but is slower to reach $k \geq 3$ ($\sim 16.9\text{k}$ vs. $\sim 3.9\text{k}$ iterations, Table 2).

Curriculum-dynamics mechanism of DSHC failure. Beyond the symmetry-group argument, the failure has a concrete curriculum-dynamics signature. When slot-permutation augmentation is disabled, slot index i is consistently occupied by ball i across an episode; because the curriculum activates balls sequentially, the slot index becomes a deterministic proxy for the current curriculum stage. The policy keys on slot identity rather than physical state, and fails to satisfy the *conjunction* of promotion criteria at $k=2$: the episode-length ratio remains high (the agent does not fail outright) but the support margin never reaches the promotion threshold. PFDS is immune by construction, since its per-frame pooling discards slot-identity information. Figure 7 reports the resulting traces in the runs we observed.

Sample-efficiency cost of disabling augmentation. Without slot-permutation augmentation, the training distribution samples only the identity element of G_{frame} at each step, reducing effective distributional diversity. Since PFDS is G_{frame} -invariant by construction, this does not prevent convergence but does increase the number of iterations required to cover the state space (Table 2: $\sim 16.9\text{k}$ vs. $\sim 3.9\text{k}$ iterations to $k \geq 3$).

F Post-Review Experimental Details

This appendix documents the additional experiments performed after the initial submission. Unless stated otherwise, all values use the paper evaluation protocol: 100 episodes per cell, $\pm 0.25\text{ m/s}$

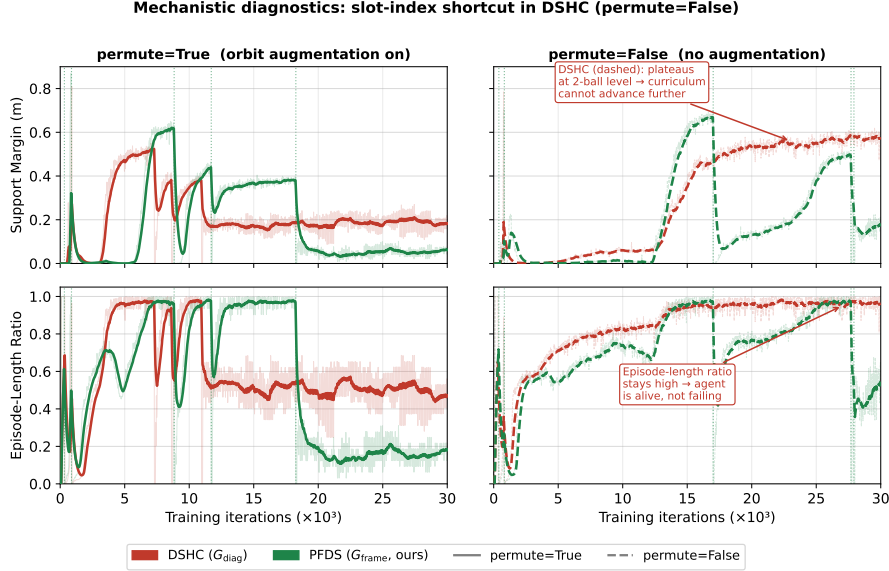


Figure 7: Curriculum-progress diagnostic for the slot-index shortcut. Rows show two promotion-related signals (top: support margin; bottom: episode-length ratio). Columns compare slot-permutation augmentation on (left) and off (right). DSHC (dashed) and PFDS (solid) are overlaid. Under augmentation-off, DSHC_OFF keeps the episode-length ratio high but cannot push the support margin to its promotion threshold, so the curriculum stalls at $k=2$; PFDS_OFF reaches both thresholds and advances. Under augmentation-on, both encoders meet the conjunction of promotion criteria.

command range, strict success as defined in Section 3, and no-drop relaxed to completion of the full horizon without a sphere leaving the plate. Additional training seeds are reported for the encoder most affected by the symmetry mismatch (DSHC_OFF) and for PFDS; a full multi-seed sweep of every augmented baseline remains future work. We therefore phrase the augmentation-free DSHC result as slow and seed-dependent, not as a universal failure. The tactile experiments isolate the representation (binary union map) and the distillation procedure (BC vs. DAgger), but do not sweep CNN/GRU architecture, history length, or resolution; those remain future work. This appendix is organized as follows: F.1 additional DSHC_OFF seeds; F.2 slot relabeling and action-level invariance; F.3 fixed- $k=5$ policies without curriculum; F.4 TACTSET robustness; F.5 DAgger versus behavior cloning; F.6 transport-quality distributions; F.7 sim-to-sim and set-size extrapolation.

F.1 Additional DSHC_OFF Seeds

Table 6: Additional DSHC_OFF training seeds under the paper setup. The submitted seed (42) stalls at $k=2$; the other seeds eventually advance, but late in training and with much lower strict success than PFDS. Evaluation uses the paper protocol (permute-on) and a matched permute-off pass. Values at $k=5$ are percent.

Seed	Curriculum	Permute on		Permute off	
		strict	no-drop	strict	no-drop
42 (submitted)	stalls at $k=2$, all 30k	—	—	—	—
41	$k=5$ at ~ 22.4 k	22	87	70	99
43	$k=5$ at ~ 15 k	23	100	79	100
PFDS (reference)	$k=5$ at ~ 6.3 k	39–41	98–100	—	—

Table 7: Test-time per-frame slot relabeling. Each cell is strict/no-drop percent, re-evaluated in the post-review harness ($n=100$). Absolute values can differ slightly from Table 1 because of simulator-asset drift; the invariant quantity is the within-encoder native-to-stress change. DSHC_OFF at $k=5$ is constant (0/0) because the submitted seed never reaches that level.

Encoder	$k=2$		$k=5$	
	native	reabeled	native	reabeled
PFDS	90/93	91/94	38/97	39/99
PFDS_OFF	98/100	96/99	24/91	42/92
DSHC	95/98	99/99	53/99	72/100
DSHC_OFF	91/94	53/81	0/0	—

Table 8: Action-level invariance gap under G_{frame} permutations, reported as a fraction of the policy’s action std (mean over 200 states $\times$ 8 permutations per sphere count; lower is better). Both models are exactly G_{diag} -invariant at numerical-noise level ($4\text{--}7 \times 10^{-4}$), so the comparison isolates their residual frame-wise dependence.

Model	$k=2$	$k=3$	$k=4$	$k=5$
PFDS	6.4×10^{-4}	5.2×10^{-4}	4.0×10^{-4}	3.6×10^{-4}
DSHC (perm. on)	2.02×10^{-1}	1.54×10^{-1}	1.23×10^{-1}	9.5×10^{-2}
Ratio	$316 \times$	$298 \times$	$307 \times$	$265 \times$

F.2 Slot Relabeling and Action-Level Invariance

F.3 Fixed- $k=5$ Policies Without Curriculum

F.4 TACTSET Robustness

F.5 DAgger Versus Behavior Cloning

The gain is concentrated in the first on-policy round and saturates thereafter, which supports a representation-limited rather than data-limited teacher–student gap.

F.6 Transport-Quality Distributions

Margin objective. The teacher reward already contains four margin-related terms: mean support margin, a CVaR $_{0.4}$ tail bonus, a near-edge penalty, and an out-of-zone penalty (Section C.1). Strengthening these terms further trades off against velocity tracking, whose error already grows with k ; a systematic weight sweep and direct distributional margin objectives are left for future work.

F.7 Sim-to-Sim and Set-Size Extrapolation

We export the distilled policy through the deployment ONNX stack and evaluate it in MuJoCo, a different contact-solver family, on a second robot variant with a larger back plate and 2 cm-radius steel spheres. The robot survives in all 250 randomized episodes ($k=1, \dots, 5$ crossed with stand, three forward speeds, and a turn command, ten seeds each). Individual sphere retention is 97.7%; 236/250 episodes (94.4%) retain every sphere, with the loss concentrated at $k=4$ (the four-ball spawn is less self-bracing than the five-ball packing); the robot itself never falls. Table 13 summarizes the per- k result. Policies trained up to $k=5$ also extrapolate zero-shot to larger sets never seen in training: the teacher retains every sphere in 98%, 95%, and 93% of $k=6, 7, 8$ episodes, while the tactile student degrades gracefully to 80%, 59%, and 33%.

G Physical Platform and Measurement Protocol

Measurement notes. The sensor publishes a pre-thresholded binary frame at 10 Hz (5 g threshold), so no continuous pressure or heat-map interpolation is available or claimed. Video and sensor streams are aligned using the bag header timestamp and the video creation time, with a +0.5 s constant

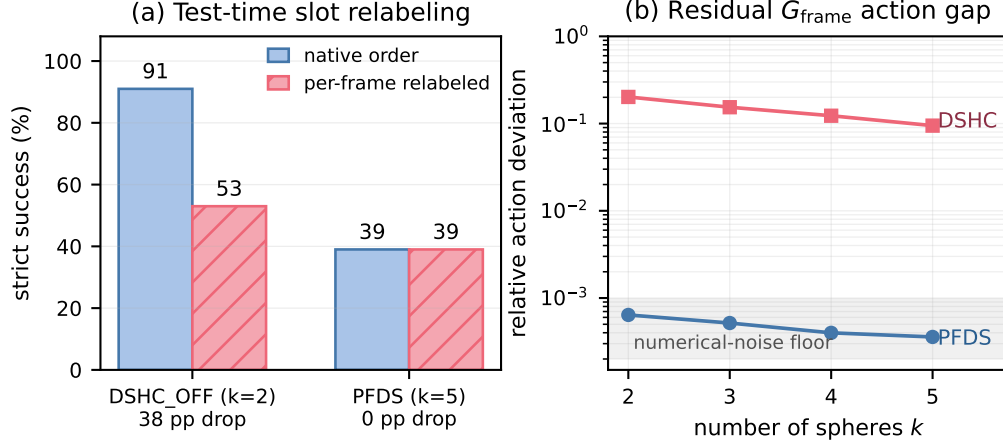


Figure 8: Symmetry diagnostics. **(a)** Test-time per-frame slot relabeling at $k=2$ reduces the submitted DSHC_OFF policy’s strict success from 91% to 53% (−38 pp), while PFDS at $k=5$ is unchanged. **(b)** Relative action deviation under G_{frame} permutations (lower is better; log scale). Both encoders are numerically G_{diag} -invariant, but augmentation-trained DSHC remains $\sim 265\text{--}316\times$ larger than PFDS, showing that augmentation yields only approximate frame-wise invariance.

Table 9: Fixed- $k=5$ specialists trained without curriculum for 15k iterations. Both encoders learn five-sphere transport, showing that the low- k curriculum plateau is not an architectural capacity limit. However, the fixed- k experts score 0% at every $k=1, \dots, 4$, whereas the curriculum policy works across all counts.

Model	$k=5$ strict/no-drop		$k=1, \dots, 4$	
	perm. on	perm. off	strict	no-drop
DSHC_OFF	73/98	70/99	0	0
PFDS_OFF	21/100	32/100	0	0

calibrated to the first ball–plate contact; the residual synchronization error is below 0.5 s, and we do not claim millisecond synchronization or taxel-level camera registration. The blob-count claim is verified by 8-connected component counting on the measured maps in selected frames. We do not report hardware success rates because repeating the same initial condition at scale is not currently feasible.

H Extensions: Equivariant Outputs and Heterogeneous Objects

PFDS is designed for a global robot action, where permutation *invariance* is the correct symmetry. For object-indexed outputs, the same per-frame construction supports within-frame permutation *equivariance*: retain each pre-pooling embedding $\phi(x_{t,i})$, pair it with the invariant frame context h_t , and decode each object’s output from $[\phi(x_{t,i}); h_t]$. This is the per-frame analogue of the Deep Sets equivariant layer [10]. Cross-frame object-indexed outputs are fundamentally different: the G_{frame} quotient identifies histories that differ by independent per-frame relabelings, so persistent cross-frame correspondence is not identifiable without an additional assumption. An explicit matching head or a temporal association module must supply that hypothesis before an equivariant decoder can produce consistent object-indexed trajectories. Global-action transport lies in the identifiable regime, which is why invariance suffices here.

For heterogeneous objects, size, mass, friction, or shape attributes enter as per-object features and break the symmetry only in the data; relabeling invariance remains correct and lossless because object identity is still not part of the observation. Separate per-type Deep Sets blocks are required only when the output itself is type-indexed, e.g., when each object type has a different action head or target assignment. The curriculum and reward remain valid for mixed-size sets as long as the

Table 10: Zero-shot TACTSET robustness on the shipped student, re-evaluated in the post-review harness. Each cell is no-drop/strict percent. The unperturbed $k=5$ no-drop rate is 85% in this harness; the originally reported 75% differs by simulator-asset generation, so all comparisons use the matched baseline.

Perturbation	$k=3$	$k=5$
None (baseline)	100/75	85/17
Taxel dropout 1%	92/59	84/13
Taxel dropout 5%	97/72	84/11
Taxel dropout 10%	95/73	84/18
Taxel dropout 20%	71/42	61/11
Spurious contact 1%	90/61	68/12
Spurious contact 5%	92/52	76/18
Calibration shift, row	87/43	86/17
Calibration shift, column	93/32	61/2
Fixed 2-frame latency	90/49	71/7
Fixed 4-frame latency	77/30	48/1

Table 11: DAgger round-by-round evaluation of intermediate students, reported as no-drop/strict percent. The round-7 row is the shipped checkpoint evaluated in the TACTSET robustness harness (the unperturbed row of Table 10).

Round	$k=1$	$k=3$	$k=5$
0 (BC only)	1/0	4/0	4/0
1 (first on-policy)	93/70	67/34	72/14
3	74/50	93/42	77/15
5	77/52	89/43	93/20
7 (shipped)	–	100/75	85/17

support-margin and termination geometry are recomputed from the per-object radius. Systematic heterogeneous experiments are left to future work.

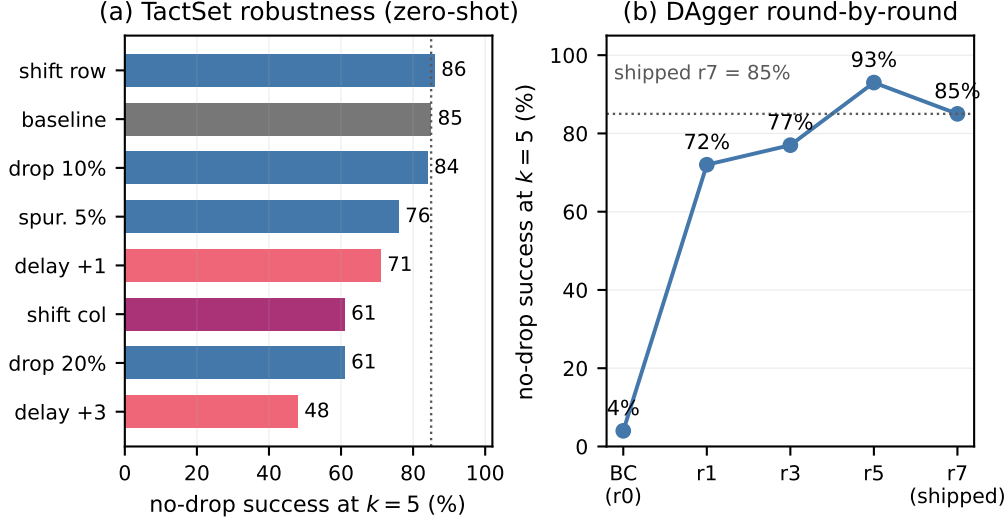


Figure 9: TACTSET robustness and DAgger. **(a)** Zero-shot no-drop success at $k=5$ under binary contact-map perturbations, sorted by severity (red: latency; purple: calibration shift); latency and column calibration shift are the dominant sensitivities. **(b)** DAgger/BC round-by-round no-drop success at $k=5$. Pure BC (round 0) reaches 4%; one on-policy round lifts this to 72%, and the shipped round-7 checkpoint measures 85% in the same harness.

Table 12: Fine-grained transport-quality statistics at $k=5$ over 100 teacher episodes (1000 steps). Margins are minimum support margin percentiles; near-edge dwell is the mean fraction of steps with a near-edge event; failure decomposition separates strict failures caused only by a transient near-edge excursion from failures that also lost a sphere.

Model	margin P10/50/90	drops	time-to-drop	near-edge	near-only / dropped
PFDS	0.000/0.000/0.061	0/100	–	0.1%	59 / 0
DSHC	0.000/0.029/0.100	2/100	90	0.4%	30 / 2
PFDS.OFF	0.000/0.000/0.085	2/100	60	0.6%	57 / 2
ST (PF)	0.000/0.009/0.182	0/100	–	0.1%	46 / 0

Table 13: MuJoCo sim-to-sim evaluation of the distilled student on the second robot variant. “Full retention” counts episodes in which every sphere remains on the plate; the robot survived in all 50 episodes per count.

Spheres	Full-retention episodes	Rate
1	50/50	100%
2	50/50	100%
3	50/50	100%
4	38/50	76%
5	48/50	96%
Total	236/250	94.4%

Table 14: Controlled physical sweep. Each run is a single qualitative closed-loop demonstration; duration is the analyzed robot-active window. “Peak taxels” and “blob count” are measured from the binary 16×16 map at the selected still frame in Figure 5.

Spheres	Duration (s)	Peak taxels	8-connected blobs
1	110.0	2	1
2	115.9	4	2
3	117.1	4	3
4	116.6	7	4
5	119.2	8	5

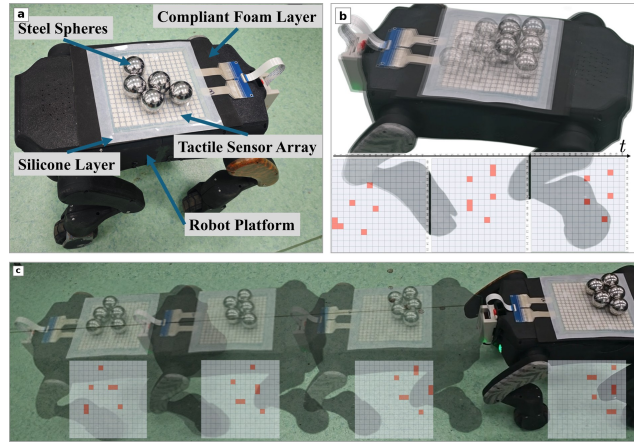


Figure 10: The physical wheel-legged platform carrying five steel spheres. (a) Sensing stack: spheres rest on a silicone cover over the 16×16 pressure array, which sits on a compliant foam layer above the robot body. (b) In-place disturbance-rejection and (c) locomotion ghost overlays of single closed-loop trials; lower panels show the binary contact maps measured by the array at the overlaid instants. The maps carry no object identity. These are qualitative demonstrations; no hardware success statistic is claimed.

I Full Per- k Success Rates

Table 15: Per- k strict/no-drop success rates (percent) for the main teacher encoders and the distilled student, evaluated with the released checkpoints and the protocol of Section 3. Teacher values use 1000-step episodes; TACTSET uses 500-step episodes.

Method	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
FLAT-MLP	0/0	0/0	0/0	0/0	0/0
BRANCH-MLP	96/96	0/2	0/0	0/0	0/0
PFDS	99/99	97/99	99/100	93/99	41/100
PFDS_OFF	99/99	99/100	90/100	77/100	41/98
DSHC	99/99	99/100	100/100	88/99	68/98
DSHC_OFF	100/100	50/78	5/13	0/0	0/0
Set Transformer	100/100	82/98	69/98	47/100	10/95
ST (PF)	98/99	97/98	93/100	82/100	54/100
TACTSET (student)	92/97	72/85	72/93	50/85	19/75

J Simulation Qualitative Rollouts

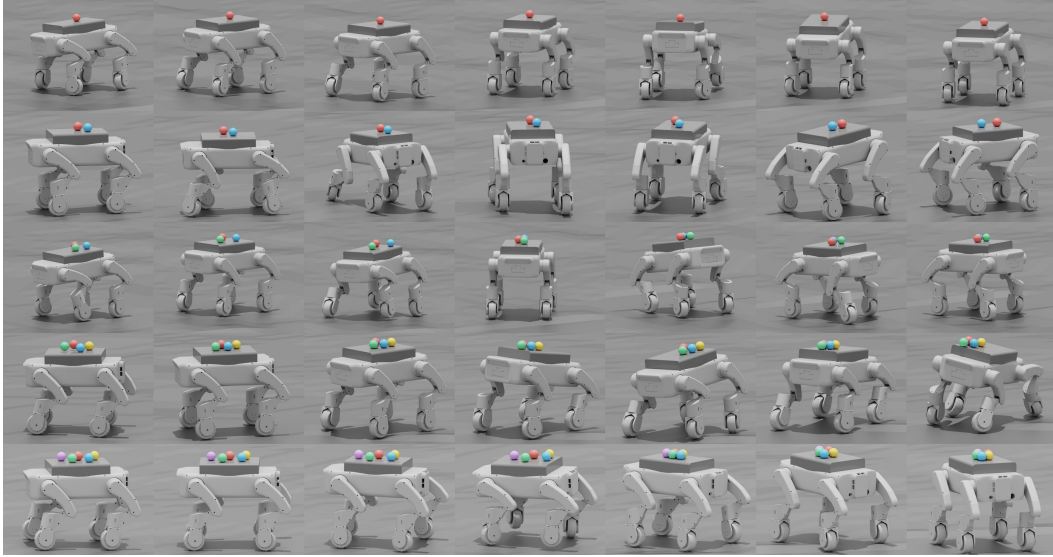


Figure 11: Simulation demonstration of multi-sphere transport with S2M-Trek. Each row shows frames from a representative episode at $k=1$ (top) through $k=5$ (bottom) balls.

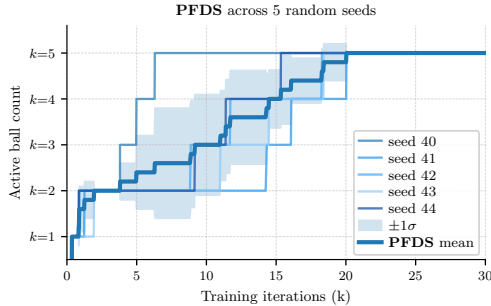


Figure 12: Multi-seed PFDS curriculum progression (mean $\pm 1\sigma$, seeds 40–44). All five seeds reach $k_{\max}=5$; the band reflects convergence-timing variation, not final performance.

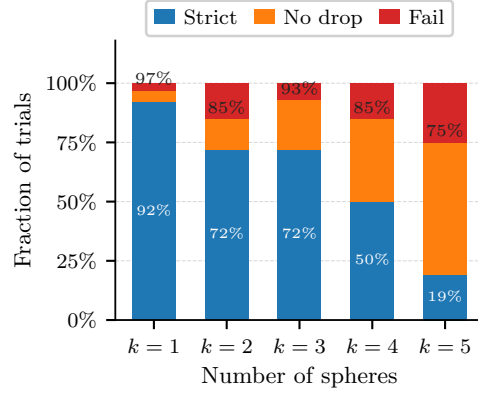


Figure 13: Original TACTSET success rates (100 trials per k , 500-step episodes; bar top: no-drop; blue inner bars: strict), matching the student results in the main text.

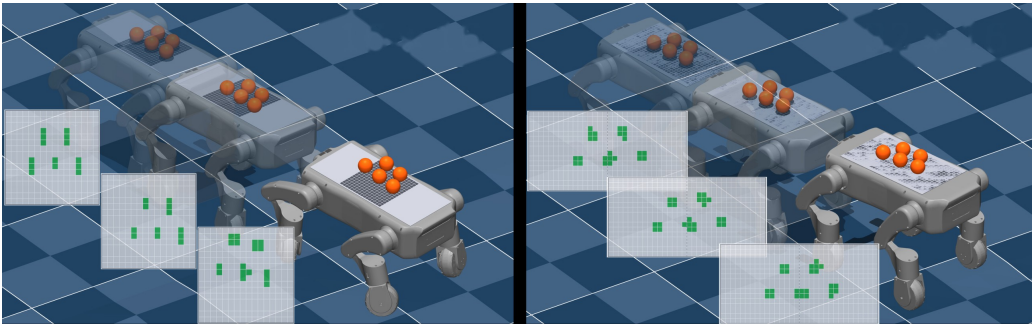


Figure 14: Closed-loop MuJoCo rollouts of the distilled student through the deployment ONNX stack on the second robot variant, carrying five spheres. Ghost trails show three timestamps of one rollout; lower panels are the binary contact maps consumed by the student.